

Planning and Reasoning Project Proposal

Giovanni Di Nola 1894682
Serena Ragaglia 1941007

January 2026

1 Introduction

For our project we wanted to create an environment where a taxi has to travel through various locations to pick up clients and leave them at their requested location. Given a series of inter-connected locations, the goal of our program is to find the least expensive way to satisfy all the requests, considering that moving through different locations can have different costs, based on whether there is a traffic light on the road or whether the location is congested.

2 PDDL domain

2.1 Types

passenger
location

2.2 Fluents

(taxiAt ?l-location): denotes that the taxi is at location l.
(userAt ?p-passenger ?l-location): denotes that passenger p is at location l.
(requestTo ?p - passenger ?l-location): denotes that passenger p has a request to go to location l.
(onTaxi ?p - passenger): denotes that passenger p is on the taxi.
(congested ?l-location): denotes that location l is congested (there is traffic).

2.3 Predicates

(road ?l1-location ?l2-location): denotes that there is a road connecting location l1 and location l2
(trafficLight ?l1-location ?l2-location): denotes that there is a traffic light between location l1 and location l2

2.4 Actions

- **move(location1, location2):**
Pre: the taxi is at location1, and location1 and location2 are connected.
Eff: the taxi is in location2 and no longer in location1.
- **charge():**
Pre: there is a user with a request in the same location as the taxi, and there is no one on the taxi.
Eff: the user is no longer in the location, but instead on the taxi.

- `discharge()`:
Pre: there is a user on the bus with a request to get down in the location where the taxi is.
Eff: the user is no longer on the taxi and the request is satisfied.

3 Situations

Three initial situations with increasing complexity will be presented to the planner. In the first one there will only be three requests, and all roads will be free and without traffic lights. In the second case the number of requests will be increased and there will be a couple of congested locations. In the third case the number of location will again increase, and there will be roads with both traffic lights and congested locations.

4 Planners and Search Heuristics

The planners we will use will be the Fast-Downward and ENSHP. We will use A* with the BLIND heuristic, h_{add} , h_{max} .

5 Controller

We will resolve a legality task to verify if it is possible to execute a sequence of actions like “*Take a passenger and bring them to this location*”. There will be a projection task to see if some condition holds after the execution of a certain sequence of actions, for example “*Passenger x at location y* ”. The controller will check if it’s possible to get to the given goal. A reactive controller will be implemented to see how the agent responds to exogenous actions, while proceeding to the goal.

The world map needs to be carefully developed in order to better define the tasks and the controller.